

How a Naked Mole-Rat Queen's Scent Keeps the Peace

From: Science News Explores Mined: 2026-10-04 5 min read

Naked mole-rats work together in colonies, but they can also fight fiercely. Scientists recently found that a queen keeps her **colony** calm with a special scent.

In a **colony**, some mole-rats look for food and others care for the young. Only the queen has babies, and she stays in charge by keeping the others from mating. When a queen dies, high-ranking mole-rats begin mating and fight hard until one becomes the new queen.

Gary Lewin is a **neuroscientist** at the Max Delbrück Center in Berlin, Germany. He studies how naked mole-rats use their senses. He had found that they use scent to tell mole-rats from other colonies apart, and he wondered whether odors matter inside a **colony** too. His team found one chemical made only by queens. It is called **isopropyl myristate**. It seems to smell very strong to the animals, because it sets off activity in the part of the brain that processes smells.

To test it, the team removed a queen from her **colony**. Then they applied the chemical to the **colony** every day for three months. "In the entire three months, there was no fighting," Lewin said, and there was no mating. When the researchers stopped adding the scent, the mole-rats began to fight, and within a few weeks a new queen emerged.

In another test, pairs of males and females were separated from the **colony**. After four weeks, no female exposed to the scent had mated. Five of six females not exposed were pregnant. Blood tests showed the chemical changes **hormones** in females, lowering **progesterone** and raising **prolactin**.

The queen makes the chemical only when she is pregnant or preparing to be. The team shared its findings on July 15 in the journal Nature.

Word Treasure

colony -- A group of animals that live together.

neuroscientist -- A scientist who studies the brain and nervous system.

isopropyl myristate -- The chemical made only by queen mole-rats.

hormones -- Chemicals in the body that send messages and control growth and behavior.

progesterone -- A hormone linked to pregnancy.

prolactin -- A hormone linked to feeding babies.

Break it down

Opening: naked mole-rats work together but fight fiercely

Colony roles and how a new queen takes over

Lewin's team finds the queen's chemical

The three-month colony test

The pair test and hormones

When the queen makes the chemical and where the study was published

Why it matters

The study shows how a single chemical can shape the behavior of a whole animal group, which helps scientists learn how animal societies keep order.

Test what you remember

1. What did scientists discover about naked mole-rat queens?
 - (A) They hide food for the whole colony
 - (B) They sing loudly to the other mole-rats
 - (C) They release a scent that keeps the colony calm
 - (D) They build new tunnels all alone
2. What does Gary Lewin study?
 - (A) How birds migrate each year
 - (B) How plants grow in dry deserts
 - (C) How naked mole-rats use their senses
 - (D) How glaciers form over time
3. What did the team do after removing the queen?
 - (A) Played recordings of the queen's voice
 - (B) Put the chemical on the colony every day for three months
 - (C) Moved the colony to a brand new lab
 - (D) Gave the colony extra food every day
4. What happened in the test with pairs after four weeks?
 - (A) All of the males left their pairs
 - (B) All of the females were pregnant after a month
 - (C) No pairs were tested during the study
 - (D) No female exposed to the scent had mated, but five of six others were pregnant
5. Which hormones did the chemical change?
 - (A) It raised progesterone and lowered prolactin instead
 - (B) It changed no hormones at all
 - (C) It lowered progesterone and raised prolactin
 - (D) It lowered only insulin in the blood
6. When does the queen make isopropyl myristate?
 - (A) Only when pregnant or preparing to be
 - (B) Every hour of the day and night
 - (C) Only after she eats a big meal
 - (D) Only when she is alone in the tunnels

Answer key (try the quiz first!): 1: C 2: C 3: B 4: D 5: C 6: A

Questions to think about

1. Gary Lewin: Said that during the three months there was no fighting, and everybody was calm with each other.

MY THOUGHTS (at least 20 words):